Light hydrogen

The hydrogen atom built from the repository's light models — toroidal-photon electron, light-ring, quark-ring or Skyrme proton, ether-knot confinement — against measured hydrogen (register entry H32)

kronspice simulation report · 2026-11-07 · recorded run: run_output.txt (about 40 s)

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Summary

Hypothesis H32 (user hypothesis, 2026-11-05) builds hydrogen from three earlier light models: the electron as the toroidal photon of Williamson & van der Mark (H23), the proton as a circuit of light (H31: one light ring, three quark rings, or Skyrme's knot), and the ether knot of H29 as what would hold the electron's light together. The binding is solved with Kron's 1945 radial Schrödinger network, both as a matrix and as a kronspice circuit whose frequency response carries the spectrum; the 21 cm line is a separate kronspice circuit of current loops. Every prediction is set beside measured hydrogen and the standard account (the Dirac electron with QED and the measured proton).

- **The rings are points at the atom's scale.** Electron ring 193.1 fm, proton light ring 0.2103 fm, Bohr radius 52918 fm (ratios 0.0036 and 4.0×10^{-6}), so the atom is the Coulomb problem with $Z = q_e q_p/e^2$.
- **The frequency response contains the spectrum.** Each level is a pole of the network's driving-point impedance; poles equal the network matrix to all ten printed digits and reach $-Z^2/2n^2$ to 3.1×10^{-6}; with q = e the Lyman and Balmer lines fall within 0.01 nm of measured, and 1S–2S is 13.4 ppm low.
- **With the toroidal-photon charge (0.910 e) the atom fails.** It carries +0.0897 e (measured limit $|q| < 10^{-21}$ e), binds 1S at 11.2686 eV and puts Lyman α at 146.701 nm (measured 121.5668 nm): every line is $1/Z^2 = 1.207$ times too long.
- **With the charge set to e the levels are Bohr's**, and the proton is tested: the light-ring proton gives the 21 cm line at 508.623 MHz (0.35808 of measured) and shifts 1S–2S by −0.907 MHz; the quark-ring proton (moment fitted) gives 1420.506 MHz but shifts 1S–2S by −0.338 MHz, against a measurement good to 10 Hz.
- **The 21 cm circuit equals Fermi's contact energy** to 1.3×10^{-5}; its ledger stored energy equals ½ΣLI² + ΣMII to 4.4×10^{-16}.
- **The confining knot predicts an unseen particle.** Sized to the electron (ℓ = 160.9 fm, energy $m_e c^2$) its squashing tone is ħω = 1.737 MeV, an excited electron at 2.248 MeV; none has been seen.

What is being tested

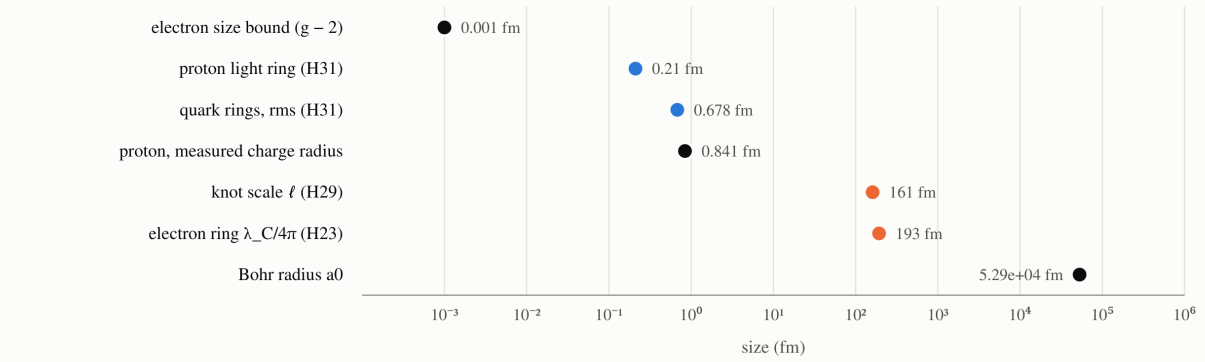
Hypothesis (H32). The hydrogen atom is the toroidal-photon electron (H23) bound to a proton of light (H31: light ring, quark rings or Skyrme knot), with the electron's light held by the ether knot (H29). **Standard prediction.** The Dirac electron with QED, bound to the measured proton, gives every measured column, including the 23.0 GHz of 1S–2S and the 0.0069 % of the 21 cm line that the Bohr and Fermi formulas leave out.

Part	Model	Numbers used
Electron (H23)	one Compton wavelength of light on a double loop	ring radius $\lambda_C/4\pi$ = 193.1 fm; charge $q = \sqrt{(3\varepsilon_0\hbar c)}/2\pi$ = 0.910 e (eq. 5); $g = 2(1 + \alpha'/2\pi)$, $\alpha' = (q/e)^2\alpha$ (eqs. 8, 12)
Electron, charge set to e	the same, with the paper's eq. 13 comparison	q = e, $g = 2(1 + \alpha/2\pi)$
Proton, light ring (H31)	+e carried at c round radius $\hbar/(m_p c)$	0.2103 fm, 1 μ_N
Proton, quark rings (H31)	uud rings fitted to the proton's moment	2.7928 μ_N, rms charge radius 0.678 fm
Proton, Skyrme knot (H31)	B = 1 hedgehog (nucleon_circuits/skyrme)	1.880 μ_N; no charge radius computed
Confinement (H29)	the ether knot	energy $547.5\sqrt{(\kappa_2\kappa_4)}$, core radius 1.2ℓ, squashing tone ω = 1.416 c/ℓ

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Fig. 6 — Sizes: both rings are far smaller than the atom

one logarithmic axis



The electron ring is 0.0036 a0 and the proton ring 4×10^{-6} a0: at the atom's scale both are points, so the atom is the Coulomb problem.

Figure 1. Sizes on one logarithmic axis: the electron size bound from g − 2 (0.001 fm), the proton light ring (0.21 fm), quark rings rms (0.678 fm), the measured proton charge radius (0.841 fm), the knot scale ℓ (161 fm), the electron ring (193 fm) and the Bohr radius (5.29×10^{4} fm). Both rings are far smaller than the atom.

The model

Kron's radial Schrödinger network

Kron's 1945 network (*Phys. Rev.* 67, 39) for s states is a radial ladder: series inductors carry the kinetic energy, shunt inductors the potential −Z/r, unit capacitors the energy, and the wave function u = rψ is the node voltage, with short circuits at r = 0 and at the edge. Eigenvalues come from a Sturm count with Richardson extrapolation to zero spacing (h → 0). The same network is built in kronspice in the dual form (capacitors and inductors swapped, because AC analysis needs a DC operating point); the network admittance vanishes at $\omega^2 = 1/(E + V0)$, so each state is a pole of the driving-point impedance. The light models change only $Z = q_e q_p/e^2$.

$Z = q_e q_p/e^2$	E_{1s} (network, h → 0)	Exact $-Z^2/2$	E_{2s} (h → 0)	Exact $-Z^2/8$
1.000000	−0.499999997 Ha	−0.500000000	−0.125000000	−0.125000000
0.910318 (H23)	−0.414339490 Ha	−0.414339492	−0.103584873	−0.103584873

kronspice twin (h = 0.1 a0, 299 nodes, AC resonance by bisection on the sign of the reactance): E_{1s} = −0.4987562112 Ha against the network matrix −0.4987562112 (difference 7.9×10^{-15}).

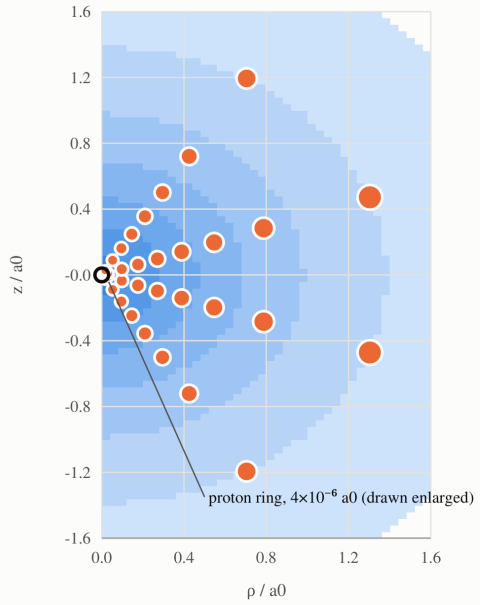
The 21 cm line as a circuit

The electron's spin, spread over the 1s cloud, is a magnetization $M(r) = \mu_e|\psi|^2\,\hat{z}$. A magnetization is a current, $J = \mathrm{curl}\,M = (2/a)M\sin\theta\,\hat{\varphi}$, so the cloud is 32 coaxial current loops round the proton ring (8 radial steps × 4 Gauss–Legendre angles, exact quadrature weights). Each loop and the ring is a kronspice inductor driven to its current, coupled to the ring by Maxwell's coaxial-circle mutual inductance. The circuit gives the aligned and anti-aligned (zz) energy ±U; spin ½ ($S\cdot S$ rather than S_zS_z) makes the splitting 4U.

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Fig. 5 — The 21 cm circuit: the electron's spin cloud as coaxial current loops

half-plane through the axis, equal scales; shading = 1s density |ψ|², dots = the 32 loops (area ∝ current)



A spin cloud is a magnetization M(r) = μ_e|ψ(r)|² ẑ.
A magnetization is a current: J = ∇ × M = (2/a)·M·sin θ φ̂,
so the cloud is a set of loops coaxial with the proton.

Each loop and the proton ring is a kronspice inductor driven to its current, coupled to the ring by Maxwell's coaxial-circle mutual inductance.

ΣM·I_p·I_k (solved currents) = μ_p·(2/3)μ0·M(0),
Fermi's contact energy, to 1.3×10⁻⁵.
The ledger's stored energy = ½ΣLI² + ΣMII to 4×10⁻¹⁶.

Loops: 8 radial steps (exact weights for e^(−2r/a)),
4 Gauss–Legendre angles (exact for sin²θ).

Figure 2. The 21 cm circuit in a half-plane through the axis: shading is the 1s density |ψ|², dots are the 32 current loops (area proportional to current), and the proton ring sits at the origin (drawn enlarged; true size 4×10^{-6} a0).

Results

The frequency response of the whole network

The whole network was swept at 8000 frequencies (1299 nodes for Z = 1, 1427 for Z = 0.910) with a 1 A source at r = 1 a0. Each pole was pinned by bisection, and two networks (spacing h = 0.1 and 0.05 a0) were extrapolated to h → 0. Energies in hartree:

n	Pole, h = 0.1	Matrix, h = 0.1	Pole, h → 0	Exact −Z²/2n²	Rel. diff
Z = 1					
1	−0.4987562112	−0.4987562112	−0.4999984496	−0.5000000000	-3.1×10^{-6}
2	−0.1249219725	−0.1249219725	−0.1249999756	−0.1250000000	-1.9×10^{-7}
3	−0.0555401320	−0.0555401320	−0.0555555534	−0.0555555556	-3.9×10^{-8}
4	−0.0312451187	−0.0312451187	−0.0312499996	−0.0312500000	-1.2×10^{-8}
5	−0.0199980004	−0.0199980004	−0.0199999999	−0.0200000000	-5.2×10^{-9}
Z = 0.910318					
1	−0.4134846441	−0.4134846441	−0.4143386084	−0.4143394918	-2.1×10^{-6}
2	−0.1035312793	−0.1035312793	−0.1035848591	−0.1035848730	-1.3×10^{-7}
3	−0.0460271288	−0.0460271288	−0.0460377201	−0.0460377213	-2.6×10^{-8}
4	−0.0258928660	−0.0258928660	−0.0258962180	−0.0258962182	-8.4×10^{-9}
5	−0.0165722065	−0.0165722065	−0.0165735796	−0.0165735797	-3.6×10^{-9}

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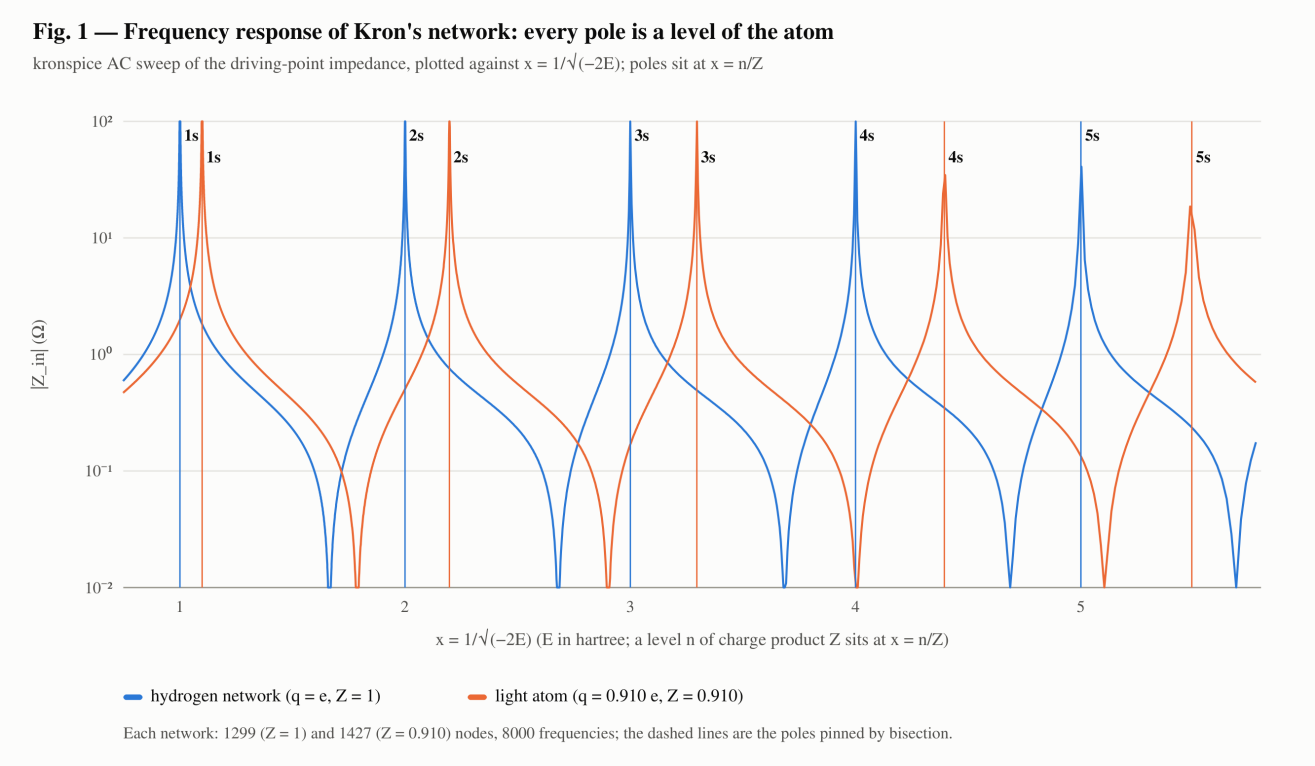


Figure 3. Frequency response of Kron's network: $|Z_{in}|$ from the kronspice AC sweep, plotted against $x = 1/\sqrt{(-2E)}$, so a level n of charge product Z sits at x = n/Z. Blue: hydrogen network (q = e, Z = 1); orange: light atom (q = 0.910 e, Z = 0.910). Vertical lines are the poles pinned by bisection.

The lines are differences of the extrapolated levels (non-relativistic Coulomb levels depend on n alone, so the s levels also fix the Lyman lines). Measured values are NIST vacuum wavelengths:

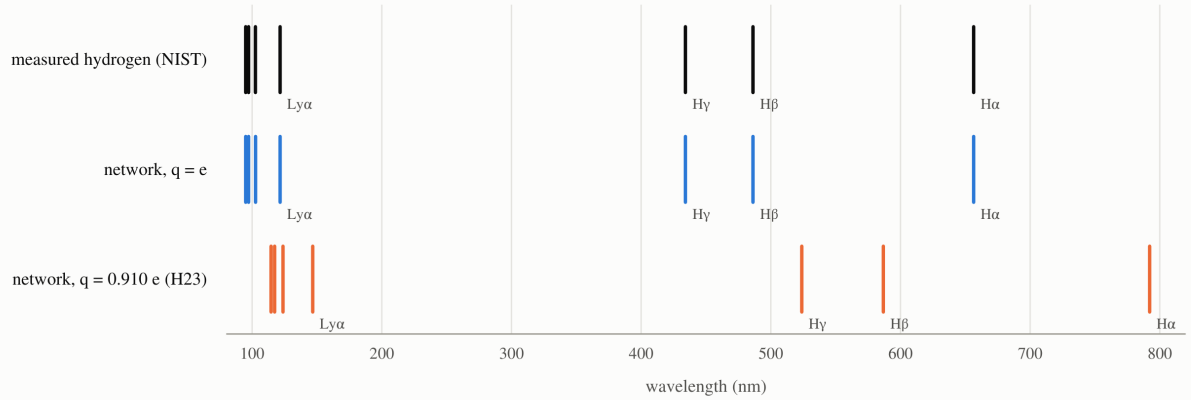
Line	Measured (nm)	q = e, from the poles (nm)	q = e − measured	q = 0.910 e (nm)
Lyman α	121.57	121.5689	−0.0011 nm	146.7019
Lyman β	102.57	102.5737	+0.0037 nm	123.7797
Lyman γ	97.25	97.2551	+0.0051 nm	117.3615
Lyman δ	94.97	94.9757	+0.0057 nm	114.6108
Balmer α (Hα)	656.46	656.4698	+0.0098 nm	792.1882
Balmer β (Hβ)	486.27	486.2739	+0.0039 nm	586.8061
Balmer γ (Hγ)	434.17	434.1731	+0.0031 nm	523.9340

1S–2S from the poles: 2466.028388 THz with q = e and 2043.548496 THz with q = 0.910 e, against the measured 2466.061413 THz (Parthey et al. 2011, ±10 Hz). With q = e the model is 13.4 ppm low, of which about 9 ppm is fine structure and Lamb shift (the Dirac and QED part) and the rest the extrapolation.

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Fig. 2 — The spectrum read from the poles, against measured hydrogen

Lyman and Balmer lines from the levels of Fig. 1 (h → 0), vacuum wavelengths



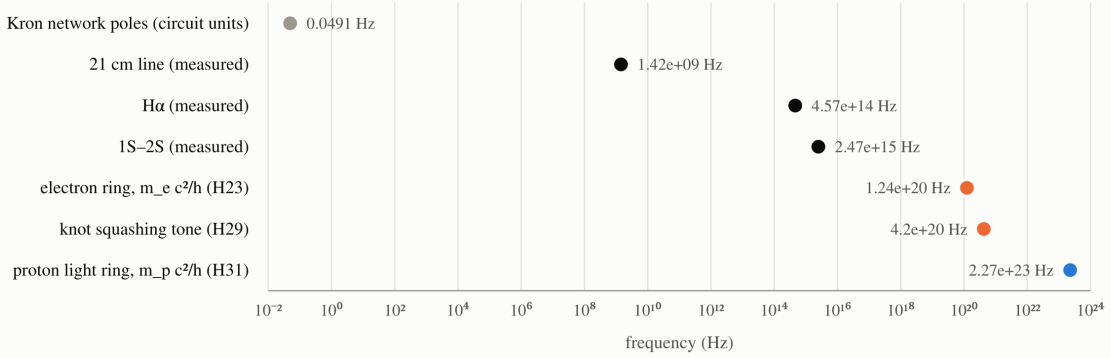
Blue lies on black to the 0.01 nm shown (Bohr levels; the 10^{-5} left is fine structure and the Lamb shift).

Orange is the same pattern stretched by $1/Z^2 = 1.207$: the light atom's lines are 21 % too long.

Figure 4. Lyman and Balmer lines read from the poles (h → 0), against measured hydrogen (NIST, black). Blue (q = e) lies on black to the 0.01 nm shown; orange (q = 0.910 e) is the same pattern stretched by $1/Z^2 = 1.207$.

Fig. 3 — Where the frequencies are: the lines are not the rings' own frequencies

one logarithmic axis



The atom's lines (black) are differences of Coulomb levels, 10^5 below the electron ring's own frequency and 10^8 below the proton's.

The network's poles are an analog: $E = 1/\omega^2 - V0$ maps them onto the levels, so their frequencies carry the levels, not the light.

Figure 5. Where the frequencies are, on one logarithmic axis: the network poles in circuit units (0.0491 Hz), the measured 21 cm, Hα and 1S–2S lines, the electron ring $m_e c^2/h$ (1.24×10^{20} Hz), the knot squashing tone (4.2×10^{20} Hz) and the proton light ring $m_p c^2/h$ (2.27×10^{23} Hz). The atom's lines come from the Coulomb binding, not from the parts' internal oscillation; the poles are an analog carrying the levels through $E = 1/\omega^2 - V0$.

Spectrum, neutrality and proton size, model by model

The levels are Bohr's with reduced mass, scaled by Z^2. The last column is the proton-size shift of 1S–2S, $(2/3n^3)(Z\alpha)^4 m_r c^2 (r/\lambda\!\!\!^{-}_r)^2$, relative to the measured radius.

Electron + proton	Net charge	1S binding	Lyman α	1S–2S size shift
toroidal photon (H23) + light ring (H31)	+0.0897 e	11.2686 eV	146.701 nm	−0.926 MHz
toroidal photon (H23) + quark rings (H31)	+0.0897 e	11.2686 eV	146.701 nm	−0.535 MHz
q set to e + light ring (H31)	0	13.5983 eV	121.568 nm	−0.907 MHz
q set to e + quark rings (H31)	0	13.5983 eV	121.568 nm	−0.338 MHz
measured (Dirac + QED) + measured	0	13.5983 eV	121.568 nm	0
Measured hydrogen	**$\lvert q\rvert < 10^{-21}$ e**	**13.5984 eV**	**121.5668 nm**	**1S–2S known to 10 Hz**

Measured: neutrality, Bressi et al., *Phys. Rev. A* 83, 052101 (2011); ionization and Lyman α, NIST ASD; 1S–2S, Parthey et al., *Phys. Rev. Lett.* 107, 203001 (2011). The 23.0 GHz between the reduced-mass Bohr 1S–2S (2466.038424 THz) and the measured value is the Dirac and QED (Lamb) part, which no row models.

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The 21 cm line

Electron + proton	μ_e (μ_B)	μ_p (μ_N)	kronspice zz energy	ν (MHz)	/ measured
toroidal photon + light ring	0.911194	1.0000	5.785×10^{-26} J	349.218	0.24586
toroidal photon + quark rings	0.911194	2.7928	1.616×10^{-25} J	975.311	0.68664
toroidal photon + Skyrme knot	0.911194	1.8800	1.088×10^{-25} J	656.529	0.46221
q set to e + light ring	1.001161	1.0000	8.425×10^{-26} J	508.623	0.35808
q set to e + quark rings	1.001161	2.7928	2.353×10^{-25} J	1420.506	1.00007
q set to e + Skyrme knot	1.001161	1.8800	1.584×10^{-25} J	956.211	0.67320
measured (Dirac + QED) + measured	1.001160	2.7928	2.353×10^{-25} J	1420.503	1.00007
Measured (Hellwig et al. 1970)				**1420.405752**	**1**

The 0.0069 % left on the standard row is the Breit, recoil and proton-structure terms beyond Fermi's formula. The quark-ring proton matches only because its moment was fitted to the measured one.

Fig. 4 — The 21 cm line from the current-loop circuit, model by model

ν / measured 1420.406 MHz; kronspice loop interaction × 4 / h

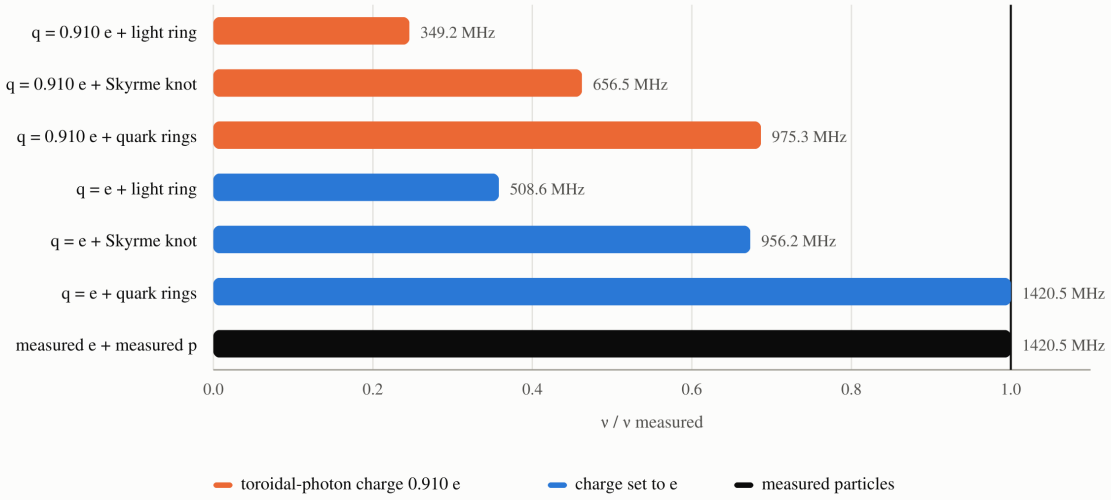


Figure 6. The 21 cm line from the current-loop circuit, model by model, as $\nu / \nu_{measured}$ (measured 1420.406 MHz; kronspice loop interaction × 4 / h). Orange: toroidal-photon charge 0.910 e; blue: charge set to e; black: measured particles.

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Fig. 7 — Light hydrogen in 3-D, at the atom's scale and at the particles' scale

left: 1s cloud (sampled from |ψ|²) with the 21 cm spin-current loops round the proton; right: the electron's double loop of light and the proton ring, true relative size

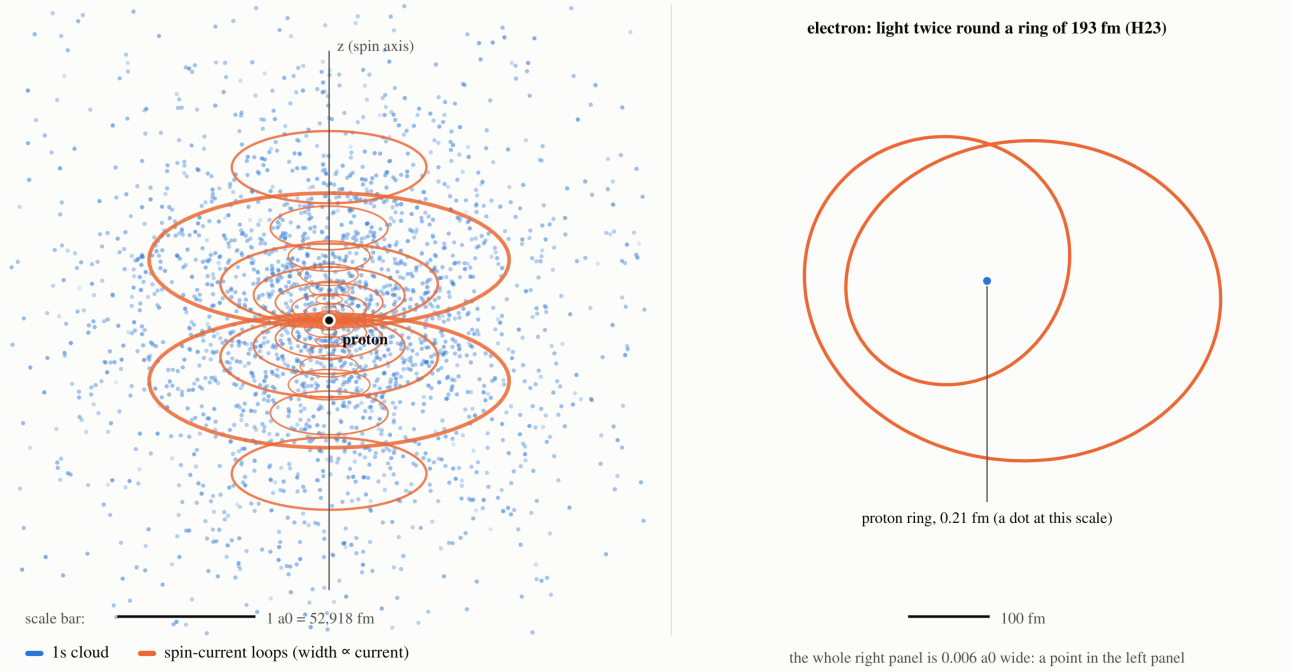


Figure 7. Light hydrogen in 3-D. Left, at the atom's scale: the 1s cloud (sampled from |ψ|²) with the 21 cm spin-current loops round the proton (scale bar 1 a0 = 52 918 fm). Right, at the particles' scale: the electron's double loop of light (ring 193 fm) and the proton ring (0.21 fm, a dot); the whole right panel is 0.006 a0 wide, a point in the left panel.

What would hold the electron's light

- **Scale:** the knot's core radius 1.2ℓ set to the ring's 193.1 fm gives ℓ = 160.9 fm.
- **Energy:** $547.5\sqrt{(\kappa_2\kappa_4)}$ set to $m_e c^2$ gives κ_2 = 0.0009294 J/m and $\kappa_4 = 2.406\times10^{-29}$ J·m.
- **Prediction:** the squashing tone ω = 1.416 c/ℓ gives ħω = 1.737 MeV = 3.40 $m_e c^2$, an excited electron at 2.248 MeV.
- **Against measurement:** collider searches exclude excited electrons to the TeV scale (PDG 2024); g − 2 limits the electron's size to about 10^{-18} m against the knot's 10^{-13} m; the knot carries no charge and no spin ½, which still come from H23's assumption.

Hydrogen under pressure and the $2p_z$ state

The network's outer short circuit is a hard wall; moving it in compresses the atom (Michels, de Boer & Bijl, *Physica* 4, 981, 1937). This is Walter Russell's "high pressure, short waves" read as the confined atom (doc/russell-octave-report.md).

Wall (a0)	E_{1s} (Ha)	E_{2s} (Ha)	E_{2p} (Ha)	Lyman α (nm)	Pressure
60.000	−0.500000	−0.125000	−0.125000	121.57	≈ 0
8.000	−0.499975	−0.084739	−0.104450	115.26	0.00157 GPa
5.000	−0.496417	+0.141254	+0.007594	90.45	0.525 GPa
3.000	−0.423967	+1.111685	+0.481250	50.36	30.1 GPa
2.500	−0.334910	+1.865480	+0.851978	38.41	97.7 GPa
2.000	−0.125000	+3.327509	+1.576019	26.80	375 GPa
1.835	+0.000219	+4.113959	+1.967265	23.18	618 GPa
1.500	+0.437018	+6.644122	+3.231051	16.32	1.93×10^{3} GPa

Selected rows of the 16 in run_output.txt (part 7).

- Compression raises every level and shortens every line; the free atom's 2s = 2p splits (2p stays lower).
- At a 2 a0 wall the ground state is exactly −1/8 Ha and Lyman α is 26.80 nm (375 GPa; the pressure at the centre of the Earth is about 364 GPa, PREM). At 1.835 a0 the ground state reaches zero: the atom is ionized by

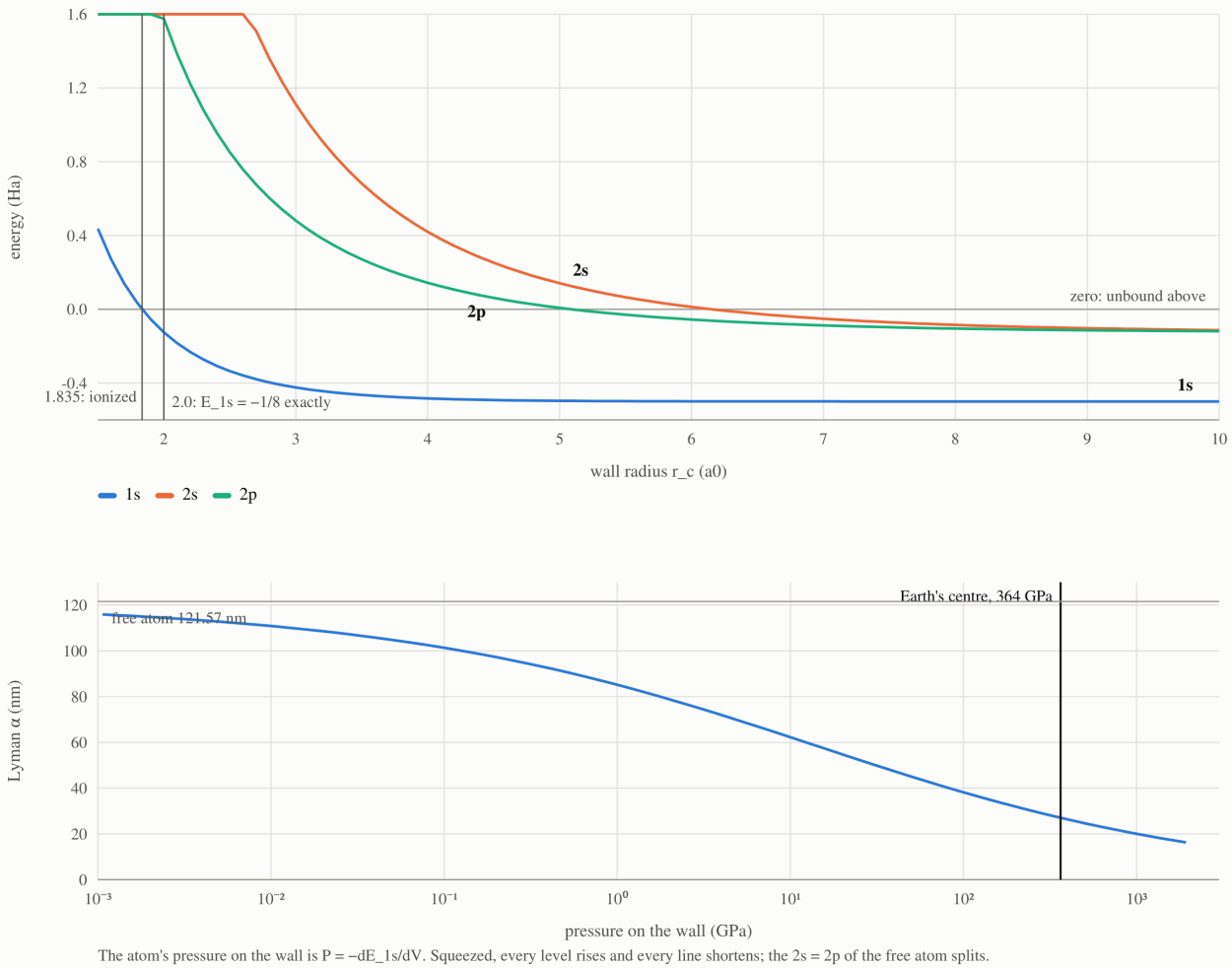
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pressure.

- With the centrifugal term $l(l+1)/2r^2$ in the shunts the network gives $E_{2p} = -0.125000000$ Ha ($h \to 0$), exact $-1/8$; its kronspice twin ($h = 0.1$ a0, 399 nodes) gives -0.125026081484 Ha against the matrix -0.125026081484 (difference 1.8×10^{-14}). The density $r^2e^{-r}\cos^2\theta$ has two lobes on the axis and is zero on the central plane.

Fig. 8 — High pressure, short waves: hydrogen compressed by a wall in Kron's network

the network's outer short circuit moved in to radius r_c; Michels, de Boer & Bijl (1937)



The atom's pressure on the wall is $P = -dE_{1s}/dV$. Squeezed, every level rises and every line shortens; the 2s = 2p of the free atom splits.

Figure 8. Hydrogen compressed by a wall in Kron's network. Top: the 1s, 2s and 2p levels against the wall radius r_c ($E_{1s} = -1/8$ exactly at 2.0 a0; ionized at 1.835 a0). Bottom: Lyman α against the pressure on the wall, $P = -dE_{1s}/dV$, from 121.57 nm for the free atom; the vertical line is the Earth's centre, 364 GPa.

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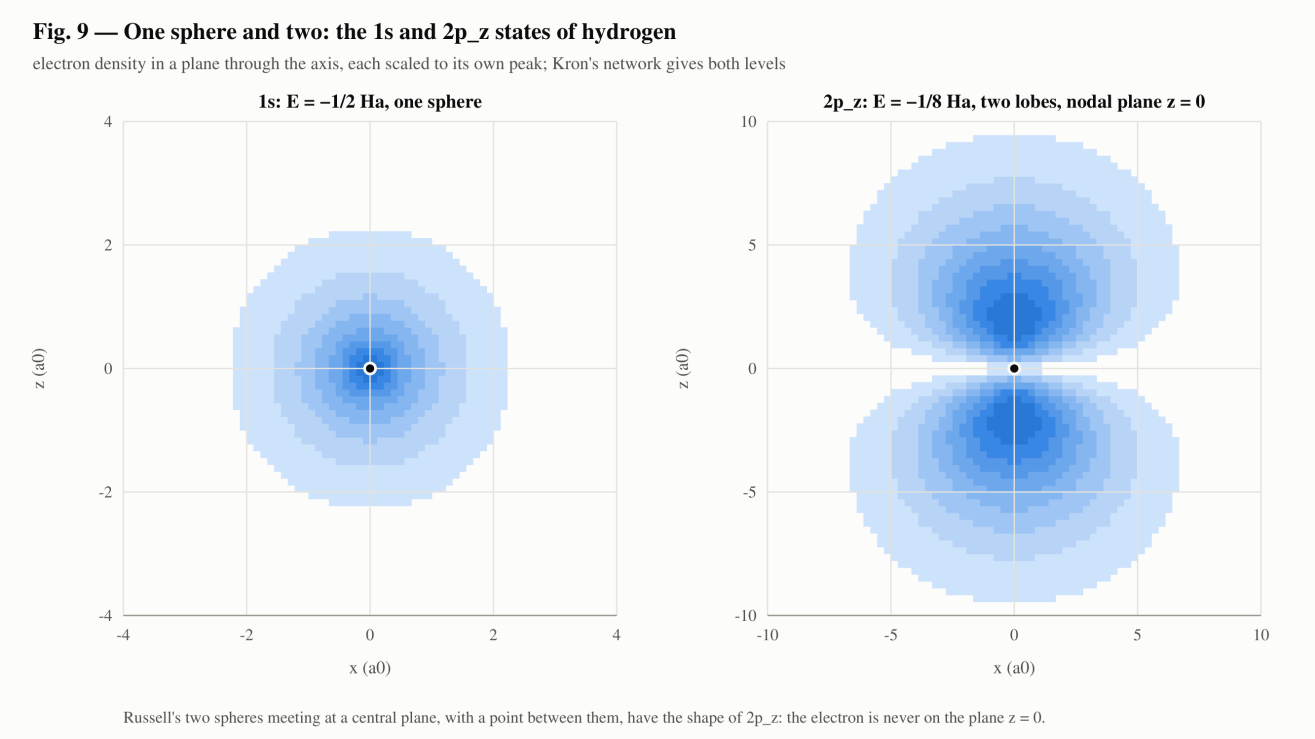


Figure 9. Electron density in a plane through the axis, each panel scaled to its own peak: 1s (E = −1/2 Ha, one sphere) and $2p_z$ (E = −1/8 Ha, two lobes, nodal plane z = 0). Kron's network gives both levels.

Validation

Test	Checks	Tolerance
TestNetworkCoulomb	network 1s and 2s against $-Z^2/2$ and $-Z^2/8$ for Z = 1 and 0.910	10^{-7}
TestNetworkTwin	kronspice resonance against the network matrix	10^{-9} Ha
TestToroidalElectron	charge 0.910318 e, g from eq. 12, ring radius 193.08 fm	10^{-6}
TestStandardHydrogen	ionization, Lyman α and the 21 cm line of the standard atom against measurement	10^{-4}
TestHyperfineCircuit	loop interaction = Fermi's contact energy; ledger = ½ΣLI² + ΣMII; 4U/h = Fermi's formula	10^{-4}, 10^{-12}
TestLightAtom	net charge 0.09 e, far above 10^{-21} e; levels scale as Z^2	10^{-12}
TestKnotExcitation	knot energy = $m_e c^2$; squashing tone 1.737 MeV	0.002 MeV
TestSpectrum	poles = matrix (10^{-9}); levels → $-Z^2/2n^2$ (5×10^{-6}); lines within 0.01 nm of measured; light-atom lines $1/Z^2$ longer (10^{-5})	as stated
TestConfined	E_{1s} = −1/8 at a 2 a0 wall; zero crossing in (1.83, 1.84) a0; 2p below 2s under compression; free 2s = 2p = −1/8	10^{-6}
TestTwin2p	the l = 1 network in kronspice = its matrix	10^{-9}
TestExport	the radial network and the hyperfine loops export, and their results sheets solve	—

Energy ledger of the 21 cm circuit: stored energy against ½ΣLI² + ΣMII, 4.4×10^{-16}; ledger residual 0.0.

Limits of the model

- The network's poles are an analog: its frequencies stand for energies through $E = 1/\omega^2 - V0$; they are not the light the atom emits.
- The levels are non-relativistic Bohr levels with reduced mass: fine structure and the Lamb shift (about 9 ppm of 1S–2S, the 23.0 GHz) are not modeled; the 21 cm circuit is Fermi's contact term, missing the Breit, recoil and proton-structure terms (0.0069 %).
- The 21 cm line is not a resonance of the network; it is a magnetic interaction energy from a separate circuit. The rings' own frequencies (1.24×10^{20} and 2.27×10^{23} Hz) are not in the spectrum.

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- The knot's constants κ_2, κ_4 are fixed here by the electron's mass and ring size, not by an independent measurement; the Skyrme proton has no computed charge radius.

Status

H32 is *bounded* in the register. With the toroidal-photon charge the atom is charged (0.090 e, excluded at 10^{-21} e), every line moves by $Z^2 = 0.829$, and the 21 cm line falls to 0.69 of the proton factor. With the charge set to e the levels are Bohr's; the light-ring proton misses 21 cm by its moment (1 μ_N against 2.793) and 1S–2S by its size; the quark-ring proton, fitted to the moment, gets 21 cm and misses 1S–2S by 0.34 MHz because it is smaller (0.678 fm against 0.841). The knot that would hold the electron's light predicts an excited electron at 2.25 MeV, not seen. The standard prediction (Dirac electron with QED and the measured proton) gives every measured column.

Reproducing the results

```
go run ./examples/light_hydrogen       # about 40 seconds
go run ./examples/light_hydrogen -svg examples/light_hydrogen/viz   # also the figures and the 3-D scene
go test ./examples/light_hydrogen
KRONSPICE_EXPORT=1 go test ./examples/light_hydrogen -run TestExport
python examples/light_hydrogen/viz/make_report.py examples/light_hydrogen/light_hydrogen_report.pdf
```

The interactive 3-D view is viz/atom3d.html. Sources: Williamson & van der Mark, *Ann. Fond. Louis de Broglie* 22, 133 (1997); G. Kron, *Phys. Rev.* 67, 39 (1945); E. Fermi, *Z. Phys.* 60, 320 (1930); Eides, Grotch & Shelyuto, *Phys. Rep.* 342, 63 (2001); Adkins, Nappi & Witten, *Nucl. Phys. B* 228, 552 (1983); Faddeev & Niemi, *Nature* 387, 58 (1997); CODATA 2022; PDG 2024; NIST ASD.

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